

6.5: L'Hôpital's Rules ; Indeterminate forms

Example :

$$\lim_{x \rightarrow 3} \frac{x^2 - 4}{x - 2} \stackrel{\text{plus } x=3}{=} \frac{3^2 - 4}{3 - 2} = 7$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \stackrel{\text{plus } x=2}{=} \frac{2^2 - 4}{2 - 2} = \frac{0}{0}$$

indeterminate form
the answer could be any
number or DNE

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2}$$

$$= \lim_{x \rightarrow 2} x + 2$$

plus $x=2$

$$= 2 + 2 = 4$$

Another way: To use L'Hôpital's rule.

L'Hospital's Rule :

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Given that : ① $f(x)$ and $g(x)$ are differentiable on
an open interval containing $x=a$, except possibly
at $x=a$

$$\textcircled{2} \quad \lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

$$\textcircled{3} \quad \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad \text{exists.}$$

This is also true when $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, $x \rightarrow -\infty$

Example :

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x^2 - 4)'}{(x - 2)'} = \lim_{x \rightarrow 2} \frac{2x}{1} \\ &= 4 \end{aligned}$$

Example : Find

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$$

check that this is indeterminate form ($\sin 0 = 0$) ✓

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \lim_{x \rightarrow 0} \frac{(\sin 2x)'}{(x)'} = \lim_{x \rightarrow 0} \frac{2 \cdot \cos 2x}{1} = 2$$

⊗ Indeterminate form of $\frac{\infty}{\infty}$

L'Hospital's Rule:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Given that: ① $f(x)$ and $g(x)$ are differentiable on

an open interval containing $x=a$, except possibly

at $x=a$

$$\textcircled{2} \quad \lim_{x \rightarrow a} f(x) = \infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \infty$$

$$\textcircled{3} \quad \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad \text{exists.}$$

This is also true when $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, $x \rightarrow -\infty$

$$\lim_{x \rightarrow \infty} \frac{3x+4}{2x+9} = \lim_{x \rightarrow \infty} \frac{3 + \frac{4}{x}}{2 + \frac{9}{x}} = \frac{3}{2}$$

use L'Hôpital's rule

$$\lim_{x \rightarrow \infty} \frac{3x+4}{2x+9} = \lim_{x \rightarrow \infty} \frac{(3x+4)'}{(2x+9)'} = \lim_{x \rightarrow \infty} \frac{3}{2} = \frac{3}{2}$$

Example : ① $\lim_{x \rightarrow \infty} \frac{9x+10}{2^x} \quad (\infty/\infty)$

$$= \lim_{x \rightarrow \infty} \frac{(9x+10)'}{(2^x)'}$$

$$= \lim_{x \rightarrow \infty} \frac{9}{2^x \cdot \ln 2} = 0$$

② $\lim_{x \rightarrow \infty} \frac{9x^4 + x^3 + x^2 + 100}{2^x} \quad (\infty/\infty)$

$$\lim_{x \rightarrow \infty} \frac{(9x^4 + x^3 + x^2 + 100)'}{2^x \cdot \ln 2}$$

$$= \lim_{x \rightarrow \infty} \frac{36x^3 + 3x^2 + 2x}{2^x \cdot \ln 2}$$

$$= \lim_{x \rightarrow \infty} \frac{(36x^3 + 3x^2 + 2x)'}{(2^x \cdot \ln 2)')}$$

$$= \lim_{x \rightarrow \infty} \frac{108x^2 + 6x + 2}{2^x \ln 2 \cdot \ln 2}$$

$$= \lim_{x \rightarrow \infty} \frac{(108x^2 + 6x + 2)'}{(2^x \cdot \ln 2 \cdot \ln 2)')}$$

$$= \lim_{x \rightarrow \infty} \frac{216x + 6}{2^x \cdot \ln 2 \cdot \ln 2 \cdot \ln 2}$$

$$= \lim_{x \rightarrow \infty} \frac{216}{2^x \cdot \ln 2 \cdot \ln 2 \cdot \ln 2 \cdot \ln 2} = 0$$

$$\textcircled{5} \quad \lim_{x \rightarrow \infty} \frac{x+1}{\ln x} \quad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{(x+1)'}{(\ln x)'} = \lim_{x \rightarrow \infty} \frac{1}{1/x}$$

$$= \lim_{x \rightarrow \infty} x = \infty$$

$$\lim_{x \rightarrow \infty} \frac{x^7 + 10x^6 + 1}{3x + \sqrt{x} + 3} = 0$$

⑤ Indeterminate Form of $0 \cdot \infty$

$$\lim_{x \rightarrow 0^+} x \cdot \ln x \quad (0 \cdot \infty)$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \quad (\infty/\infty)$$

$$= \lim_{x \rightarrow 0^+} \frac{(\ln x)'}{(1/x)'} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2}$$

$$\left[\text{note: } \left(\frac{1}{x}\right)' = (x^{-1})' = -1 \cdot x^{-1-1} = -x^{-2} = -\frac{1}{x^2} \right]$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{x} \cdot (-x^2) = \lim_{x \rightarrow 0^+} -x = 0$$

Try:

$$\lim_{x \rightarrow 0^+} x \cdot \ln x = \lim_{x \rightarrow 0^+} \frac{x}{1/\ln x}$$